Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



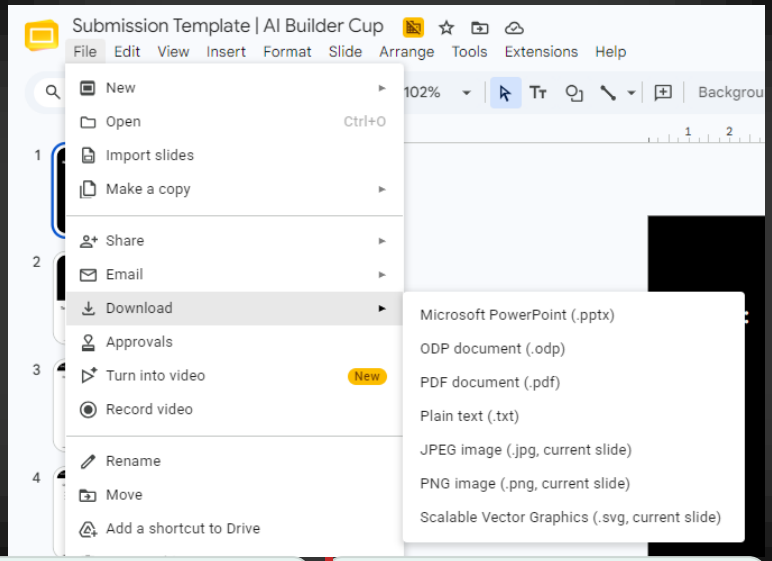
TEAM DETAILS · AI BUILDER CUP 2026 · JAPAC · SUSTAINABILITY

Team name: Agent Ram

Team leader: Ramasamy P

Product: TerraSentinel AI — ground-motion early warning

Problem: After each Sentinel-1 pass, operators face thousands of InSAR points with no consequence ranking. TerraSentinel cuts motion into 250 m sites, ranks who inspects today (RPI), reports honest time-to-threshold bands (never failure dates), and uses Gemini to draft a number-verified inspection ticket.



TerraSentinel AI

Which 250 m should an engineer inspect today?

Engine ranks consequence. Gemini writes the ticket. Every number checked.

87 → 7
sites → act list

RPI 86
NH-58 · 35–64 d band

Gemini
verified tool numbers

GCP
Run · BQ · Vertex

BRIEF ABOUT THE IDEA

Ground creeps for months. Nobody ranks which creep matters.

Mountain highways · NH-58
Sole valley access. One slip cuts towns & supply.

Metro tunnels · Chennai
Settlement under roads & buildings. Tight tolerance.

Dams & tailings
Crest / toe motion with severe downstream consequence.

The decision we own
Which 250 m needs an engineer, why, and how soon?

Today's options fail the decision
- Raw InSAR portal — no asset link, no who-goes-today
- Rainfall alarm — trigger treated as hazard
- Generic AI — invents mm and failure dates
- Walk-over calendar — ignores acceleration now

We do
- Creep on linear assets
- Rank · explain · track
- Threshold window + assumptions

We refuse
- Failure / collapse dates
- Sudden inter-pass events
- Certified safety claim

OPPORTUNITIES · USP

Not an InSAR viewer — a decision layer with honest outlooks and guarded GenAI.

How different?

- 250 m sites on corridor geometry
- RPI = hazard x consequence x confidence
- Alerts = pass diffs; Act list = today
- Rain amplifies only (+50% cap)
- USGS quakes: context, never scored
- Gemini narrates; engine decides

How it solves it

- Validate → locate → measure
- WEAK guard: bad radar != CRITICAL
- Rank Act-today shortlist
- Outlook = ETA band, never a date
- Ticket: summary · cause · checklist
- Same maths: local XOR BigQuery

USP / demo value

- Honest continuing-regime bands
- Number guard rejects invented mm
- Production GCP posture (Run + BQ)
- 62,330 obs → 87 sites → 7 action
- 4 Act today in Chamoli
- Judge can sign in and act live

FEATURES

Every surface ships a decision — not a dashboard toy.

Command Center

Map + KPIs + Act today, ranked by consequence

Site drawer

RPI, speed, rain, outlook band, pass trend

Alerts

What changed between passes

Brief · Trends · Portfolio

Planning views; history recomputed from data

Work Orders + Gemini

Template-first; 7 tools; number guard

Data Hub

Validate-then-import; rain fetch; provenance; BQ sync

Access & audit

RBAC · sessions · CSRF · audit log

Methodology

Published formulae — every score auditable

PROCESS / USE-CASE

One satellite pass: space → ranked site → crew at the right 250 m.

1 Pass
Every 6–12 days

Sentinel-1 mm motion

2 Map
Onto the asset

1,355 pts → 87 sites

3 Rank
What matters

Speed · accel · consequence

4 Gemini
Write ticket

Tools → guard → checklist

5 Next
Keep watching

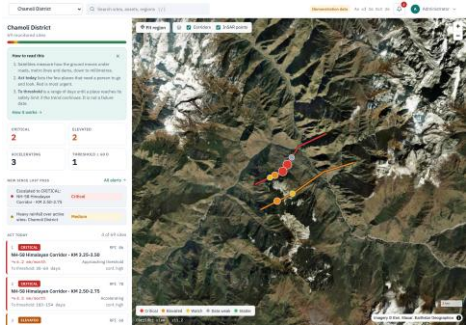
Act · next pass replay

Demo funnel (labelled synthetic)

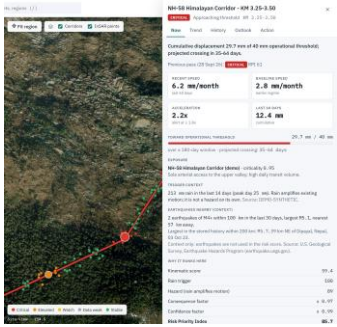
62,330	1,355	87	7	4
observations	radar points	250 m sites	CRIT/ELEV	Act today

PRODUCT UI (LIVE)

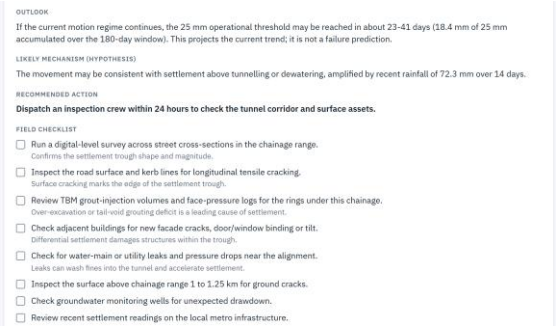
Running product screens — aspect-correct, not stretched wireframes.



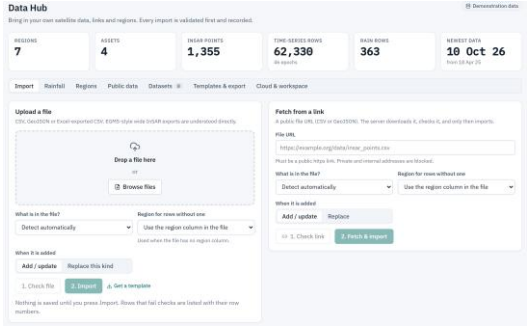
Command Center · Act list



Site · RPI & outlook



Gemini work order



Data Hub · validate then import

ARCHITECTURE

One container · two storage backends · same tested maths.

EDGE	Cloud Run · HTTPS · CSP/HSTS · rate limits · sessions · GCS state
UI	React 19 · MapLibre · React Query · role-aware screens
API	FastAPI · RBAC · CSRF · SSRF-safe fetch · audit
ENGINE	numpy · pandas · shapely — RPI, tiers, ETA bands, alerts, pass replay
GENAI	Gemini tool loop · number guard · model fallback · Vertex / AI Studio
DATA	BigQuery (GEOGRAPHY, partitions) XOR local · USGS · provenance

TECHNOLOGIES

Google Cloud end-to-end. Gemini turns a risk score into an action.

Ingest	Store	Compute	Explain	Act
Cloud Run Data Hub	BigQuery GIS + history	Cloud Run risk engine	Gemini Vertex/Studio	Engineers map + ticket

Gemini — what GenAI does

- 7 tools: metrics, history, outlook, rain, seismic, exposure, quality
- Must call tools first → investigate → JSON → number guard
- Rejects invented mm and collapse language
- Fallback flash-latest → lite on 429/5xx; template always ships

GCP services

- Vertex AI / AI Studio — Gemini
- BigQuery — partitions, ST_DWITHIN, bytes-billed cap
- Cloud Run — one image: SPA + API + engine + agent
- Secret Manager · GCS · IAM via ADC (no key files)

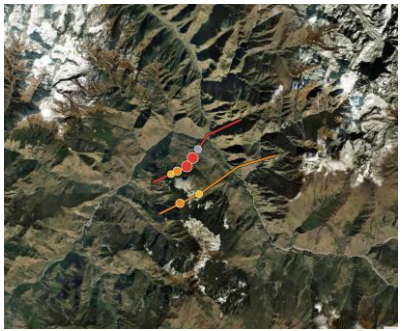
ESTIMATED COST (OPTIONAL)

GenAI only on tickets; ranking stays deterministic CPU.

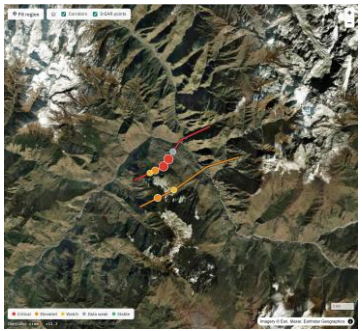
Cloud Run	1–2 vCPU / 2 GiB · min=1 while judging	low tens USD/mo warm
BigQuery	~62k rows · TTL cache · max bytes billed	pennies–low USD demos
Gemini Flash	Work-order path only · template fallback	per-ticket, optional
Secrets + GCS	Admin/Gemini secrets · state volume	pennies
Scale path	Materialize assignment · raise instances	linear with corridors

PROTOTYPE SNAPSHOTS

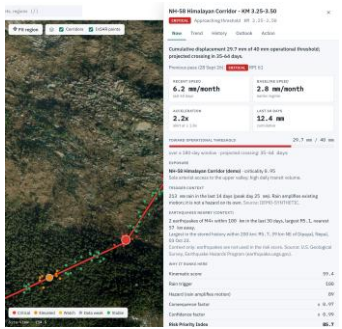
Demo value on screen — Chamoli, NH-58 CRITICAL, guarded ticket.



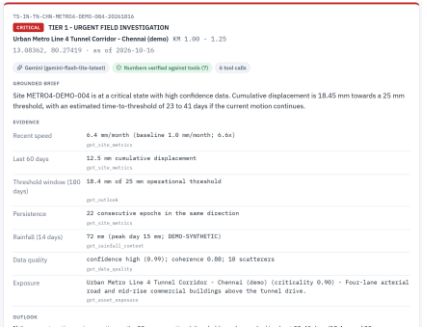
Chamoli · 69 sites · 4 act today



Ranked map — not every scatterer



NH-58 · RPI 86 · 35-64 day band



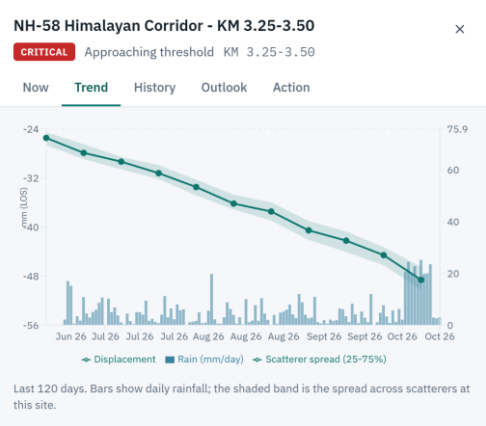
Gemini · numbers verified

PERFORMANCE / BENCHMARKING

NH-58 KM 3.25-3.50 — same numbers in UI, tools, and work order.

Worked example · CRITICAL · RPI 86

- Inputs: 6.2 vs 2.8 mm/mo · 2.2x · 29.7/40 mm · rain 213 mm
- kin = 100*(.4*0.62+.3*0.40+.3*0.74) = 59.4
- hazard = 59.4*(1+0.5*1.0) = 89.1 (rain capped)
- RPI = 89.1*0.975*0.985 ~ 85.7 → CRITICAL
- Outlook 35-64 days if trend continues — NOT a failure date
- History: site 61→86 · action sites 3→7 over 16 passes
- What-if: minor road 57 ELEV · weak radar ≤15 · rain-only ~1



Site trend · displacement + rain

Live Cloud Run + BigQuery (warm)

summary ~0.4s · sites ~0.3s · alerts ~2s

Hot-path prefetch before history warm

GenAI never computes RPI / tier / ETA

FUTURE DEVELOPMENT

Scale the data plane — honesty rules stay fixed.

Roadmap

- Production InSAR on the same Data Hub schema
- Materialize point-assignment for national scale
- HA once sessions leave single-process files
- Field checklist / offline crew packet
- OIDC / IAP for enterprise buyers
- Deeper rain/geology as amplifiers only

Buyer · ask · honesty

- Buyer: asset-integrity / geotech teams
- What they buy: attention — 1,355 pts → 7 sites
- Ask: live demo + 3-min video + repo
- Outlook bands are never collapse dates
- Gemini narrates tools only
- Synthetic demo data labelled in UI

SUBMISSION LINKS

GitHub · 3-minute demo · live product.

GitHub public repository

https://github.com/ramasamyp-aiengineer/terra-sentinel-ai

Demo video (3 minutes)

docs/pitch/TerraSentinel-AI-Demo-3min.mp4 (upload Drive/YouTube for portal)

Final product (live)

https://terrasentinel-598899050820.asia-southeast1.run.app

Sign-in credentials shown on the login page · Judge walk: docs/JUDGE.md

RISK ENGINE MATHS

Measure → Score → Project. Deterministic, tested, explainable.

1 · Measure

- Recent speed: slope last 60 d
- Baseline: 270 to 90 d ago
- Accel = |recent|/max(|base|,0.5)
- Persistence · seasonal recovery
- Site = median in 250 m bin
- conf mixes coh·epochs·pts·SNR
- Weak data can never be CRITICAL

2 · Score

- kin=100*(.4s+.3a+.3p)
- hazard=kin*(1+0.5*rain/200)
- RPI=hazard*(0.5+0.5*crit)
- *(0.5+0.5*conf)
- Slow/far x0.2 · seasonal x0.35
- WEAK cap 15 · quakes never scored
- CRITICAL / ELEVATED / WATCH

3 · Project

- ETA_const = remaining / speed
- ETA_accel = quadratic root
- band=[min,max] <= 365 d
- Highway 40 mm / 180 d demo
- Metro 25 mm · Dam 30 mm
- ALWAYS a band
- NEVER a failure date

Google Cloud H2S

TerraSentinel AI

Thank you — Team Agent Ram

Which 250 metres should an engineer inspect today?

The engine ranks them. Gemini writes the ticket. Every number is checked.

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Live demo + GitHub + 3-min video

Google Cloud AI Builder Cup 2026 · Sustainability

Gemini · Vertex AI · BigQuery · Cloud Run · Sentinel-1